Predicting Adolescent Victimization, Perpetration, and Their Overlap: A Dual-Classifier Analysis Using Global Predictors from a Representative Violence Survey in Adolescents

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Author contributions statement

All authors contributed to the study conception and design. Material preparation and data collection were supervised by NP and MB. Formal analyses were performed by OP and IG. The original draft was written by OP and IG. All authors discussed the results and commented on the manuscript. The review and final manuscript were written by NP and MB.

Ethical considerations

The study adhered to the ethical principles outlined in the Helsinki Declaration on research involving human subjects (World Medical Association, 2013). Ethical approval for the research was obtained from the ethics committee of the University of Barcelona (IRB00003099).

Comentado [NP1]: SI PREDECIMOS LOS TRES ROLES ESTE DEBE SER EL TÍTULO, SI NOS ENFOCAMOS EN EL OVERLAP, HAY QUE CAMBIARLO POR ESTE OTRO TÍTULO: Disentangling Victim–Perpetrator Overlap: A Dual-Classifier Analysis Using Global Predictors from a Representative Violence Survey of Adolescents

Comentado [NP2]: TOT EL QUE ES PUGUI PASSAR A SUPPLEMENTAL MATERIALS S'HA DE PASSAR. EN EL TEXT CAL DEIXAR NOMÉS L'INDISPENSABLE PER A COMPRENDRE LES ANÀLISIS I ELS RESULTATS.

Comentado [NP3R2]: He reduit les futures línies d'estudi i eliminat les dues taules del final.

Comentado [NP4R2]: Entenc que totes les variables s'han de traduir a l'anglés a les taules, o potser no?

Comentado [MOU5R2]: Jo ho traduiria, pero a veure que opinen la resta

Con formato: Sangría: Izquierda: 0 cm, Sangría francesa: 1,27 cm

Consent to participate

All participants provided written informed consent prior to participation. Because all participants were 14 years or older, according to Spanish law, parental consent was not required for participation in this anonymous school survey. No financial assistance or compensation was provided to participants.

Declaration of conflicting interest

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of the article.

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Data availability statement

The data that support the findings of the study are available in the OSF repository, but restrictions apply to the availability of the data, which were used under license for the study, and so are not publicly available until 2026. Data ~~are~~is, however, available from the authors upon reasonable request.

Abstract

Objectives: This study examines the overlap between adolescent victimization and perpetration using a dual-classifier framework based on global predictors. The objective is to develop an interpretable, recall-oriented screening tool capable of identifying adolescents at heightened risk of being victims, perpetrators or the overlap of both roles at the same time.

Methods: Analyses were conducted on a representative sample of 4,024 school-aged adolescents in Spain (14–17 years old), with data collected online through the eAlicia platform. To address class imbalance, the victim–perpetrator overlap was decomposed into two balanced binary classification tasks: (i) victimization and (ii) perpetration. Victim status was modelled with a cost–complexity pruned decision tree, and perpetrator status with a compact feed-forward neural network. Predictors were processed through a pipeline of feature selection, engineering, and scaling, followed by orthogonalization via principal components analysis (PCA). Each classifier was trained on these components, and an ensemble rule predicted dual-role status only when both models concurred.

Results: The victim classifier achieved a recall of 91.1% (F1 = 0.68), while the perpetrator classifier reached 90.0% recall (F1 = 0.45). The ensemble, though more conservative, preserved high sensitivity with a recall of 93.8% and achieved a balanced accuracy of 70.3%. Compared with single models, it substantially improved specificity, reducing false positives and enhancing utility for screening in prevention-focused settings.

Conclusions: The dual-classifier framework offers a transparent, high-sensitivity approach to assessing violence risk. The decision tree enables rapid identification of potential victims, with human interpretable results. ~~while~~ At the same time, the neural network captures behavioral volatility linked to offending, but with limited human interpretability. Shared predictors—particularly household structure, substance use, and coping resources—emerge as key mechanisms underlying the victim–perpetrator overlap.

Keywords: victim–perpetrator overlap; dual classifier; decision tree; neural network;

Comentado [NP6]: REVISEU perquè l'he tocat i no vull dir res que no hàgim fet.

Comentado [MB7]: En principi és màxim 250 paraules però diuen a les guidelines que es poden fer exceptions. Ara en tenim 280.

Comentado [NP8]: ¿ES ESTE EL OBJETIVO O PREDECIR LOS TRES PERFILES?
JO DIRIA QUE ELS TRES PERFILS...

Comentado [O9R8]: Modificado con sugerencia Marina

Comentado [MB10]: Si fem dels tres jo diria: of being victims, perpetrators or both.

Comentado [O11R10]: Sí perfecte.

Comentado [O12]: No superar las 250 palabras

Comentado [O13]: 4-6 palabras clave

~~1.~~ INTRODUCTION

Victimization and offending are both highly prevalent during adolescence, and empirical research increasingly demonstrates that these roles often intersect (Cuevas et al., 2007). Victimization is widespread among juvenile offenders (Pires & Almeida, 2024), and many young people who experience relational victimization also engage in relational aggression (Casper et al., 2020). Similarly, adolescents involved in dating violence frequently move fluidly between victim and perpetrator roles (Park & Kim, 2019). This convergence of roles reflects one of the most striking findings in criminology—the victim–offender overlap, where individuals may simultaneously experience victimization and engage in offending (Jennings et al., 2012). Despite this evidence, most quantitative studies continue to treat victimization and offending as separate outcomes (Turanovic, 2022), which makes it more difficult to identify the common underlying causes they share and, in turn, ~~obscures shared mechanisms and~~ limits integrated crime prevention strategies. Social and institutional discourses also reinforce this divide, classifying individuals as either victims or offenders and overlooking the “grey zone” in which both roles coexist. In this sense, research indicates that s~~S~~tudying victimization and offending in isolation provides only a partial view of the phenomenon, overlooking the interconnected nature of these experiences (Berg & Mulford, 2020). ~~Social and institutional discourses also reinforce this divide, classifying individuals as either victims or offenders and overlooking the “grey zone” in which both roles coexist.~~By contrast, i~~I~~gnoring the overlap and ~~l~~labeling a young person solely as an offender risk missing key opportunities for intervention and can lead practitioners to overlook their victimization. In parallel, labeling a young person solely as a victim risks neglecting their agency and the factors driving their own offending behavior (XXX).

~~The association between offending and victimization weakens with age (Schreck et al., 2017), making adolescence a critical period for examining how these roles intersect and for identifying those at risk of occupying both. Studying victimization and offending in isolation provides only a partial view, overlooking the interconnected nature of these experiences (Berg & Mulford, 2020).~~ Adopting a victim–offender overlap framework moves beyond this fragmented perspective toward a more comprehensive understanding of violent behaviors (van Gelder et al., 2015). This approach can reveal modifiable factors that prevent adolescent victims from becoming perpetrators. At the same time, adolescence is a critical period for examining how these roles intersect and for identifying those at risk of occupying both, since the association between offending and victimization weakens with age (Schreck et al., 2017). ~~By contrast,~~

Comentado [MB14]: Això m’ho he inventat una mica però em sembla que serveix per justificar més l’estudi d’overlap.

Con formato: Resaltar

~~ignoring the overlap and labeling a young person solely as an offender risk missing key opportunities for intervention and can lead practitioners to overlook their victimization.~~

Addressing this complexity has fueled calls to shift from description to prediction (Rosenbusch et al., 2021), emphasizing methods that anticipate, rather than merely document, adolescent outcomes. Machine learning models, for example, outperform traditional statistical techniques in forecasting phenomena such as suicidal thoughts and behaviors (Weller et al., 2021), extremist attitudes (Haghish et al., 2023), bullying perpetration (Zhou et al., 2025), and school violence risk (Ni et al., 2020). Beyond predictive accuracy, these methods help mitigate measurement bias stemming from underreporting of sensitive experiences (Chen et al., 2023). Applying predictive approaches to adolescent victimization, perpetration, and their overlap offers a promising pathway for the early identification of at-risk youth, the implementation of effective trauma-informed interventions, and the design of more targeted prevention strategies.

~~2.~~ BACKGROUND

Adolescent violence is a complex, multidimensional phenomenon that represents a major challenge for public health worldwide. Both victimization and offending are highly prevalent during this developmental period. Meta-analyses have shown that violence is a common experience during adolescence, with more than one in three young people worldwide reporting exposure to at least one form of violence—physical, sexual, or emotional—within the previous year (Hillis et al., 2016; Stoltenborgh et al., 2015). Evidence from international surveys underscores that adolescents are not only victims but also active participants in violent behaviors. For instance, Gatti et al. (2011) documented that 16% of adolescents aged 12 to 17 in a sample spanning 30 countries acknowledged engaging in violent acts over the course of a year. Similarly, Enzmann et al. (2010), drawing on data from 31 European nations, observed self-reported prevalence rates exceeding 20% among 12- to 15-year-olds.

A critical feature of adolescent violence is the overlap between victimization and offending (Zun et al., 2005). Research consistently shows that young people who engage in offending are more likely than their peers to have histories of victimization (Fitton et al., 2020), while victims face heightened risks of violent behavior (Peltonen et al., 2020). For instance, Erdmann and Reinecke (2018) found in a longitudinal study of German students aged 14 to 20 that, at age 14, 8.5% of youth fell into the victim–offender overlap category. This represented 40.0% of all victims who also offended and 44.4% of all offenders who also experienced victimization. While the victim–offender overlap has been observed across the life course (Lauritsen & Laub, 2007), adolescence is the stage when this association is most pronounced (Beckley et al., 2018).

Taken together, these findings emphasize adolescence as a pivotal stage for the emergence and entanglement of violent behaviors and victimization. Understanding these dynamics is crucial not only for refining theoretical models of violence but also for informing prevention strategies that can reduce both the immediate and long-term harms associated with youth violence and victimization.

~~2.1.~~ Risk Factors for Adolescent Victimization and Offending

Adolescent victims and offenders frequently share overlapping characteristics, life experiences, and behavioural patterns. In fact, many of the risk factors that predict delinquency - such as family problems, substance abuse or poor school performance - also predict victimization (Fagan & Mazerolle, 2011; Farrington et al., 2012). Furthermore, victimization has been found as a significant risk factor for involvement in delinquency (Pires & Almeida, 2024; Shaffer & Ruback, 2002). However, relatively few studies have investigated which factors specifically differentiate victim–offenders from those who are exclusively offenders or exclusively victims (Beckley et al., 2018). Clarifying both the distinctions between these groups and the risk factors common to both roles is essential for advancing our understanding of youth violence and for designing more effective prevention strategies during adolescence (Farrington et al., 2017). Importantly, progress in this area requires moving beyond the analysis of individual or contextual variables in isolation, and instead examining how these factors interact and reinforce one another (Trinidad et al., 2018). Although existing research confirms a strong association between victimization and offending, deeper insights are still needed into the unique vulnerabilities of victim–offenders and the mechanisms that heighten their risk for involvement in both roles (Ousey et al., 2011).

Several theoretical frameworks have been proposed to explain victimization, offending, and their overlap (Berg & Schreck, 2022). Among the most influential are the exposure perspective, which includes Hindelang et al.'s (1978) lifestyle theory and Cohen and Felson's (1979) routine activity approach, the control theory (Gottfredson & Hirschi, 1990), and the social interactionist–situational perspective (Berg &

Comentado [MB15]: Veig que just surt més abaix aquests exemple. Es pot canviar per sexual promiscuity.

Comentado [O16R15]: Em sembla bé.

Con formato: Resaltar

Felson, 2020). These perspectives are especially relevant during adolescence, a developmental stage marked by profound changes in both violent behavior and vulnerability to victimization that extend from childhood into adulthood (Erdmann & Reinecke, 2018). Empirical research consistently shows that victims and offenders share comparable sociodemographic and psychological characteristics (Berg & Mulford, 2020).

The exposure perspective emphasizes how adolescents' daily routines and lifestyle choices shape opportunities to encounter risky situations, thereby increasing the likelihood of both offending and victimization. Some risk factors operate similarly for boys and girls, while others differ, as daily routines can vary considerably by gender (Pusch & Holtfreter, 2021). Age is another relevant dimension: although the risk of delinquent behavior peaks in mid-adolescence and tends to decline thereafter, this pattern appears to reflect lifestyle differences shaped by social and cultural changes rather than being solely the result of a fixed biological trajectory (Steffensmeier et al., 2025).

From the perspective of control theory, individual traits such as impulsivity and other self-regulation deficits explain part of the association between victimization and offending. Longitudinal research shows that low self-control is a common feature of trajectories of both delinquency and victimization (Jennings et al., 2010), while meta-analytic evidence confirms that low self-control is a robust predictor of both crime (Pratt & Cullen, 2000) and victimization (Pratt et al., 2014).

Other variables relevant to control theory include moral emotions. Guilt and shame occupy a prominent place in criminological research, as both have been shown to inhibit criminal behavior, with meta-analytic studies confirming their protective role (Spruit et al., 2016). In particular, self-reported conscience among young people—comprising empathy, guilt, and moral reasoning—has been found to reduce the likelihood of delinquent behavior (Walters, 2022), whereas moral disengagement is consistently associated with aggressive conduct in childhood and adolescence (Gini et al., 2014). In this sense, Falla et al., 2020 suggest that victims of violence may disengage morally through cognitive restructuring, normalizing aggression and, in turn, becoming more prone to reproduce the violence they experienced. Importantly, moral emotions also appear to function as mediators linking trauma (DeCou et al., 2023) and victimization (Whiffen & MacIntosh, 2005) to the development of psychopathology. Thus, the integration of moral emotions into the study of the victim–offender overlap may help explain why experiences of victimization and offending often co-occur.

Beyond individual traits, contextual factors such as family instability—characterized by insufficient parental guidance and inconsistent monitoring—have also been linked to delinquent behavior (Kierkus & Hewitt, 2009) and victimization (Turner et al., 2013). These processes are closely tied to the absence of social support, long recognized as a central element in offending (Cullen, 1994) and more recently suggested as a potential factor underlying victimization as well (Azimi & Daigle, 2020).

From a social interactionist–situational perspective, expanding the routine activities, lifestyles, and social control frameworks to incorporate additional exposure variables—such as alcohol use, pornography consumption, and running away behavior—offers a valuable avenue for understanding the victim–offender overlap (Turanovic & Pratt, 2014).

Alcohol intoxication is a well-established correlate of aggression and criminal behavior, as it escalates conflict by increasing hostility, reducing risk awareness, and weakening adherence to social norms and precautionary strategies (Felson et al., 2008). In a cross-national study of adolescents from 30 countries, Gatti et al. (2011) found that alcohol use was reported by 25.4% of participants, but the prevalence rose to 63.1% among those embedded in antisocial peer groups, underscoring the role of social context. While the link between alcohol and adolescent aggression is well-documented (Willhelm et al., 2020), meta-analytic evidence also demonstrates that alcohol consumption heightens vulnerability to victimization—though this relationship remains less studied than its connection to offending (Duke et al., 2018). Notably, this association is observed in both genders but tends to be stronger among males (Popovici et al., 2012).

Pornography consumption represents another controversial exposure variable. Reviews have struggled to establish a consistent relationship between pornography use and sexually violent behavior (Ferguson & Hartley, 2022), though some evidence suggests violent pornography may intensify sexually aggressive conduct (Wright et al., 2016). Importantly, adolescents often interpret pornographic images as realistic, shaping their emotional and cognitive perceptions of sexuality (Gunnoo & Powell, 2023). Empirical studies reinforce these concerns: in U.S. samples of youth aged 16–17, weekly pornography use has been associated with increased risk of physical or sexual victimization in dating relationships (Rothman

& Adhia, 2016), while longitudinal data show that adolescents aged 10–15 who view violent pornography are nearly six times more likely to later report sexually aggressive behavior (Ybarra et al., 2011).

Running away also represents both a consequence and a predictor of risk. It is frequently a response to victimization, particularly in cases of physical or sexual abuse (Sundin & Baguley, 2015), but simultaneously increases the likelihood of engaging in criminal behavior (Jeanis et al., 2019). Youth homelessness more broadly is strongly associated with both victimization and violent offending (Heerde & Hemphill, 2019), and running away is closely tied to sexual exploitation (Franchino-Olsen, 2021). Thus, running away operates as a bidirectional risk factor, linked both to experiencing violence and to subsequent antisocial or criminal conduct (Kaufman & Widom, 1999).

Quantitative Approaches to the Victim-Offender Overlap

Currently, most studies have approached the victim–offender overlap from a descriptive perspective. In their effort to understand the phenomenon, researchers have typically focused on identifying shared correlates of victimization and offending rather than developing models capable of predicting which adolescents are most likely to occupy both roles (Jennings et al., 2012). Predictive approaches based on interpretable decision-tree models and compact neural networks are advantageous because they generate individualized risk estimates and capture complex relationships among predictors (XX). Rather than estimating average associations, they produce individualized risk scores that reflect the joint configuration of multiple global predictors for each adolescent (XXX). This enables the modeling of complex interactions and non-linear relationships that underlie victimization, perpetration, and their overlap (XXX). When implemented within an interpretable framework, these models also make it possible to identify the relative contribution of key predictors to risk (XXX). This makes it possible to identify which individuals are most likely to benefit from particular programmes and tailor interventions (XX). At the same time, algorithms can uncover long-range, cross-context and socially structured patterns in victimization and offending, and outperform traditional statistical methods in predicting violence, recidivism, and serious crime (Berk et al., 2009; Berk & Hyatt, 2015). In addition, ML approaches make it possible to explicitly weight false positives and false negatives according to ethical and policy priorities, improving the alignment between predictive aims and the social consequences of error (XX). Finally, AI systems can be audited to identify where bias emerges and why a particular output was produced (XXX).

Comentado [MB17]: Tot això és nou, a veure què en penseu…

Comentado [MB18R17]: Help Georgina, segur que tu això ho saps molt millor…

Comentado [MB19R17]: Bé, això ho puc revisar i acabar de redactar, però primer m'agradaria que em diguessis si creu que vai bé Georgina i si saps articles/referències per sustentar-ho també aniria molt bé…

Comentado [MB20]: Això és el que van dir, almenys, l'Ignasi i l'Oriol

Comentado [MB21]: Això també ho van dir al debat sobre IA i Crimi que vaig fer

...there remains a need for quantitative frameworks that both forecast the overlap and preserve transparency in how predictions are generated.

~~2.2. Current Study~~The Present Study

Building on this foundation, the present study uses artificial intelligence techniques to model and predict adolescent victimization, adolescent offending, and the combination of both as victim–offender overlap.~~,~~ Overall, the system allows to ~~aiming to~~ identify ~~those~~ adolescents in urgent need of intervention due to any of the three behaviours mentioned before. For each participant, we analyze the most recent complete calendar year of observation, examining behavioral, environmental, and psychosocial indicators to identify correlates of victimization and offending and to explore patterns associated with their overlap. The analysis is cross-sectional and descriptive. To prevent label leakage, items directly defining victim or perpetrator status are excluded from the predictors.

~~and explore mechanisms underlying their overlap.~~ To address class imbalance, we implement a dual-classifier framework that decomposes the overlap into two balanced binary tasks. The pipeline includes: (i) a three-stage feature-preparation procedure to systematically select, engineer, and orthogonalize predictors; (ii) two high-recall classifiers—a cost-complexity pruned decision tree for victimization and a compact feed-forward neural network for offending; and (iii) an ensemble logic that labels adolescents as victim–perpetrators only when both models concur. This approach provides a principled decomposition of a tri-label problem, enables assessment of global predictors shaping both roles, and offers an end-to-end tool for evidence-based screening and triage.

The study ultimately ~~aims to enhance~~ provides a tool to help in the early identification of adolescents who, as victims, perpetrators, or both, are at heightened risk and in urgent need of support. ~~By providing a~~This high-sensitivity screening framework~~, it~~ informss timely interventions in educational, clinical ~~school~~ and residential settings, and can help reduces the likelihood of continued involvement in violence~~, and promotes safer, more resilient communities.~~ In doing so, ~~it~~ it helps build safer communities and equips practitioners with evidence-based tools to respond more effectively to the complex needs of youth involved in victimization, offending or both.~~advances evidence-based strategies for prevention and triage, helping practitioners address the complex needs of youth occupying dual roles and fostering environments that mitigate risk and support well-being.~~

Comentado [MB22]: Afegim objectius o passem?

Comentado [NP23R22]: De moment hem de TALLAR TEXT. Potser sí que caldria posar-los...

Comentado [O24]: Hay que modificar esto:
- Se habla de "mecanismos", pero el diseño es transversal y no causal; no infiere causalidad.
- Se cita la polivictimización, pero no se modela: se excluyen ítems definitorios de víctima/perpetrador para evitar leakage.

Comentado [MB25R24]: He modificado el texto para reflejar esto pero valdría la pena revisarlo de nuevo.

~~3.~~ METHODOLOGY

Please notice, t~~T~~echnical details of the method and results sections can be found in the ~~online~~ supplemental materials.

Comentado [NP26]: Passar el màxim que es pugui a supplemental materials.

~~3.1.~~ Participants and Survey Dataset

Con formato: Sangría: Izquierda: 0 cm, Sangría francesa: 1,27 cm

The study adopts a cross-sectional, classificatory, and non-causal design. All indicators refer to the same 12-month reference window and are analyzed contemporaneously, with the aim of detecting statistical regularities that predict victimization, perpetration, and their overlap, without inferring temporal ordering or causality.

Analyses were based on an anonymous survey of adolescents enrolled in secondary education centers in Spain (ages 14–17). A stratified cluster sampling design was used, with strata defined by type of education (Compulsory Secondary Education, High School, and Vocational Training~~ESO, Bachillerato, Basic Vocational Training, and Intermediate Training Cycles~~), school ownership (public/private), and ~~autonomous community~~region.

The final sample comprised n = 4,024 participants: 2,061 girls (51.2%), 1,858 boys (46.2%), 36 identifying with another gender (0.9%), and 69 with no response (1.7%); mean age M = 15.42 (SD = 1.03). Most were born in Spain (n = 3,654; 90.8%); 339 (8.4%) were of foreign origin; the remainder did not report. Regarding living arrangements, 95.8% (n = 3,856) lived with their father, mother, or both.

~~3.2.~~ Measures

To ensure interpretability, each item ~~was~~ is assigned to one of four mutually exclusive categories: Global, Perpetrator-defining, Victim-defining, and Derived targets. Only Global variables (demographic, psychosocial, lifestyle, and contextual factors that do not directly define the outcomes) ~~entered~~ are used in the predictive models. Perpetrator-defining and Victim-defining items ~~were~~ are used exclusively to construct the Derived targets (victimization, perpetration, and their overlap) and ~~were~~ are never ~~included~~ used as variables in the predictors.

Comentado [O27]: He eliminado la referencia a la tabla de la categorización de variables, dado que la información que nos aporta ya esta reflejada en dicho parrafo.

Global predictors. Thirty-five variables T**able 1** that after conceptual filtering and recoding, **27 engineered** predictors ~~entered~~ are used in the analysis pipeline (coded so that higher values indicate greater risk/vulnerability; see **Table 2** for labels, type, and descriptives). These include ~~sociodemographics~~sociodemographic variables (gender, age, country of birth, household composition), psychosocial dispositions (impulsivity; general self-efficacy; social support; moral emotions), lifestyle exposures (alcohol, pornography), and running away from home/residential centers. All Global variables were used as predictors only to avoid label leakage.

Perpetrator-defining. A list of 24 binary (0/1) items covering violent/antisocial behaviors in the past 12 months (conventional offenses, violence against parents, peer aggression, sexual perpetration/exploitation, electronic aggression). These were excluded as predictors because they define the perpetration construct.

Victim-defining. Thirty-two binary (0/1) items assessing past-year victimization (Spanish adaptation of the JVQ), expanded with sexual exploitation and online victimization. These were excluded as predictors because they define the victimization construct.

Derived targets. From the above sets we defined three outcome variables: victimization (y^v) = any endorsement among the 32 victim items; perpetration (y^p) = any endorsement among the 24 perpetration items; and overlap (y^{vp}) = $y^v \wedge y^p$. Count totals from each battery were retained for descriptive purposes and did not enter as predictors.

~~3.3.~~ Data Processing Pipeline

The raw corpus consists of a cross-sectional survey matrix of the form $D \in R^{nxp}$ $with\ n = 4024$ adolescent respondents (i.e., matrix's rows) and $p = 347$ items spanning demographic, socio-cultural and behavioral domains (i.e., matrix's columns). The matrix has been pre-processed via ~~Listwise~~list-wise deletion of 6.4 % incomplete cases yields a processed ~~working~~ matrix of $n = 3767$ complete observations. Before modeling a three-stage, pipeline was applied:

1) Data Quality Assessment & Curation (DQAC);
2) Feature Engineering & Rebalancing (FER);
3) Normalization & Latent Compression (NLC)~~:~~

Comentado [MOU28]: Mejor que entered

Comentado [O29]: Esta sección parece dividida en dos bloques (survey data y measures). Propongo integrarlos en un único apartado y simplificar los detalles técnicos del procesamiento. Podemos dejar esos detalles completos referenciados en *Supplemental Materials.* ¿Os parece?

Comentado [MB30R29]: Justamente esta estructura del artículo es bastante clásica e informadora. Si hay que eliminar partes igual no sé si lo haría de aquí.

~~3.3.1 Data Quality Assessment & Curation (DQAC)~~

~~Operations.~~

focused on transforming the selected variables into consistent, risk-oriented, and comparable predictors while addressing class imbalance.

sequence of transformations applied to produce the final 27-element feature set. Categorical variables for gender and sexual orientation were one-hot encoded to avoid artificial ordinality (Poslavskaya, 2023). Age values were clipped to the 14–17 range, matching the valid limits of the questionnaire. Rare categories in residential-care variables were merged to ensure that each binary variable represented a sufficiently large subgroup.

For multi-item constructs - Impulsivity (8 items), Self-efficacy (5), Social Support (7), and Moral Emotions (5) - the mean, median, and variance were computed. Items written in a protective direction were reverse-coded so that higher scores consistently indicated greater risk. This step produced twelve continuous predictors, reducing dimensionality, mitigating multicollinearity, and facilitating principal components analysis (PCA). Pornography exposure was aggregated by recording the highest reported frequency across items.

To handle missing data, unanswered substance-use questions were recoded as high risk (value = 1), and all other missing values were also set to 1 to reflect informative missingness. A final verification ensured that all engineered variables increased monotonically with the hypothesized likelihood of victimization or offending, resulting in a coherent 27-variable vector.

1. ~~**One-hot encoding**: GENERO.BN and ORIENTSEX.BN are expanded into two binary columns (i.e., with only two possible values each) each to avoid spurious ordinality. Ekaterina Poslavskaya(28 December 2023)~~

Comentado [MOU31]: Oriol: Busca una referencia que explique esto a profanos de la estadística.
Creo que esta puede servir, pero mira si te gusta:
Shachar Kaufman; Saharon Rosset; Claudia Perlich (January 2011). "Leakage in data mining: Formulation, detection, and avoidance". *Proceedings of the 17th ACM SIGKDD international conference on Knowledge discovery and data mining*. Vol. 6. pp. 556–563. doi:10.1145/2020408.2020496. ISBN 9781450308137. S2CID 9168804. Retrieved 13 January 2020.

Comentado [O32R31]: Veo bien esta referencia para explicar el leakage.

Con formato: Fuente: Sin Negrita, Cursiva

Con formato: Fuente: (Predeterminada) Times New Roman, 10 pto

Con formato: Fuente: (Predeterminada) Times New Roman, 10 pto

Comentado [MB33]: Esto es así? Qué significa en este contexto "high risk"?

Comentado [I34]: TBD @Oriol : Añadir cita Poslavskaya, Ekaterina, and Alexey Korolev. "Encoding categorical data: Is there yet anything'hotter'than one-hot encoding?." *arXiv preprint arXiv:2312.16930* (2023).

Comentado [I35R34]:

~~3.3.3~~ *Normalization & Latent Compression (NLC)*. The Normalization and Latent Compression (NLC) stage aimed to make all variables comparable, remove collinearity, and concentrate the predictive information into a reduced set of orthogonal factors. Equations (2) and (3) describe the normalization and projection processes applied to the engineered variables. By projecting the data onto the first 27 principal components, the procedure retained the largest share of variance, eliminated multicollinearity, and substantially reduced computational cost, as summarized in Table 3.

~~**Objective.** Render variables comparable **(eq.2)**, remove collinearity, and concentrate predictive signal into orthogonal factors. **(eq.3)**~~

Table 3.

~~3.4. Model Building~~Model Development and Classification Framework

The modelling strategy was designed to balance two complementary objectives: maximizing sensitivity to both victimization and offending, while maintaining a level of simplicity that allows interpretation and use in frontline settings. All model development and validation were conducted in collaboration with the eAlicia AI research team. To address the severe class imbalance inherent in the victim–perpetrator overlap, the task was decomposed into two binary classification problems—victimization and perpetration—whose outputs were subsequently combined through a logical ensemble rule. Victimization was modelled using a decision tree trained on the first 18 principal components, which together accounted for approximately 95% of the total variance in the engineered predictor space. To control variance and ensure transparency, the tree was subjected to cost–complexity pruning, resulting in an extremely sparse model consisting of a single split. Model selection was performed via grid search over the pruning parameter, yielding an optimal value of $\alpha = 0.01495$. This configuration retained one threshold on the second principal component (PC2) at 0.54, producing a fully transparent, rule-based classifier. Despite its simplicity, the model achieved a recall of 0.91 for the victim class on the hold-out set, ensuring that the vast majority of victimized adolescents were correctly identified.

Perpetration was modelled using a compact feed-forward neural network trained on the first 22 principal components, capturing approximately 99% of the variance. The architecture was intentionally constrained, comprising roughly 2,000 trainable parameters, to balance expressive capacity with

Comentado [MB36]: Esta tabla igual se podría poner en resultados, porque qué componentes concentran la mayor parte de la varianza es parte de lo que hemos descubierto ¿no?

Comentado [O37R36]: Sí que podría ponerse en resultados y de hecho con un mapa de temperatura que es más visual que los propios valores.

Comentado [O38]: Me gusta la subdivisión en tres secciones. Propongo reducir el número de bullets, mantener el contenido a un nivel más alto y trasladar los detalles técnicos a *Supplemental Materials*.

Comentado [MB39R38]: Me gusta la estructura. He hecho una propuesta que es necesario revisar. La idea es eliminar información de esta sección, así que todo lo que penséis que se puede acortar, adelante.

generalization and interpretability. The network consisted of two hidden layers (64 and 8 units, respectively), with dropout applied during training, and a sigmoid output producing a probabilistic estimate of perpetration risk. Class imbalance and overfitting were addressed through inverse-prevalence class weighting, dropout regularization, early stopping based on validation recall, and an explicit overfit-gap criterion. Hyperparameters were selected via grid search, prioritizing the simplest architecture capable of achieving the predefined recall target. On the hold-out set, the network attained a recall of 0.90 for the perpetrator class.

Dual-role (victim–perpetrator) cases were identified using a conservative AND-ensemble that flagged an individual as overlap only when both the victim and perpetrator classifiers returned positive predictions. This conjunction preserves high sensitivity while substantially reducing false positives, yielding a parsimonious and fully transparent detection rule. Formally, overlap status was assigned only when both binary outputs were equal to one. This strategy avoids the instability that would arise from directly training a model on the overlap label, which is characterized by extreme class imbalance, and instead leverages two better-balanced learning problems whose conjunction isolates the highest-risk cases. The resulting ensemble maintains interpretability, stabilizes optimization, and is well suited for screening and triage in applied settings, where missing a high-risk adolescent carries substantially greater cost than investigating a false alarm.

Model Explainability (XAI)

Model explainability was addressed for each component of the framework to ensure that predictions could be traced back to the original global predictors. For the victim classifier, the cost–complexity–pruned decision tree reduces to a single split in principal-component space, governed by a threshold on the second principal component (PC2). Because the model consists of only one decision rule, its behavior is fully transparent. Contributions of the original predictors were recovered by algebraically back-projecting the PC-based split into raw-variable space, inverting preprocessing transformations and normalizing coefficients to obtain unit-free importance weights. This procedure yields, for each predictor, a direction of effect and a relative contribution to victim classification, which are reported in Table 10.

Explainability for the perpetrator classifier was derived using a two-step procedure that preserves the neural network's non-linear structure while enabling interpretation at the level of the original predictors. First, Shapley values were computed for each principal component to quantify their global contribution to the network's output. These component-level importances were then propagated back to the engineered predictors using the PCA loading matrix, producing signed variable-level contributions and normalized importance shares. The resulting rankings of principal components and original predictors are reported in Tables 11 and 12, respectively.

For the overlap classifier, feature importance was derived by integrating the explanations of the two base models combined through the logical AND rule. When a predictor appeared in both models, its direction and weight were reconciled by comparing contributions across classifiers: concordant effects were summed, while discordant effects were netted, retaining the dominant direction. Final weights were rescaled to sum to one. The resulting ensemble-level importances, presented in Table 13, preserve the interpretability of the decision tree while incorporating the broader signal captured by the neural network, yielding a transparent account of the predictors [illegible]

~~**3.5. Model Explainability (XAI)**~~

Con formato: Fuente: Sin Negrita

Con formato: Fuente: Sin Negrita

Comentado [MOU40]: Ejemplo de que vale más una imagen que mil palabras.

Comentado [MOU41]: Referenciadlo

Comentado [O42R41]: Wythoff, B. J. (1993). Backpropagation neural networks: A tutorial. *Chemometrics and Intelligent Laboratory Systems, 18*(2), 115–155. https://doi.org/10.1016/0169-7439(93)80052-J

Comentado [MOU43]: Añadid referencia

Comentado [O44R43]: Gewers, F. L., Ferreira, G. R., de Arruda, H. F., Silva, F. N., Comin, C. H., Amancio, D. R., & da F. Costa, L. (2021). Principal component analysis: A natural approach to data exploration. *ACM Computing Surveys, 54*(4), Article 70. https://doi.org/10.1145/3447755

Comentado [MOU45]: ¿Podemos poner una imagen sencilla que lo explique, como la de arriba?

Comentado [MOU45]: ¿Podemos poner una imagen sencilla que lo explique, como la de arriba?

Con formato: Color de fuente: Automático

How to read the metrics (Table 5). We prioritise recall—the share of true positives detected—because missing a victim or a perpetrator carries the highest social cost in contrast to for instance, classifying a non-victim as victim. Precision gauges frontline workload (what fraction of alarms are correct). Specificity indicates how many low-risk youths are correctly left unflagged; together with recall it forms balanced accuracy, which remains informative under class imbalance. We also report accuracy (used only as a rough check), F1 (the precision–recall harmonic mean), and NPV—critical here because it tells practitioners how safe it is to trust a "negative" decision.

Victim detection (decision tree). The depth-1 CCP-pruned tree identified 339 of 372 true victims (recall 91.1%), so fewer than one in ten victims were missed. At the same time, TN=95, FP=287, FN=33, TP=339, which yields precision 54.2% (about half of the flagged youths are truly victims), specificity 24.9% (intentionally sacrificed to privilege sensitivity), accuracy 57.6% (not a reliable signal under imbalance), NPV 74.2% (three in four "safe" calls are truly non-victims), and F1 67.8% (confirming practical usability despite recall-oriented tuning). These figures align with the screening purpose: accept more false alarms in exchange for very few misses.

Perpetrator detection (neural network). The compact NN retrieved 199 of 221 perpetrators (recall 90.0%), again surpassing the sensitivity target. Counts were TN=245, FP=476, FN=22, TP=199, giving precision 29.5% (roughly one in three alarms is correct), specificity 34.0%, accuracy 47.1%, NPV 91.8% (a strong guarantee that negatives are truly safe), and F1 44.5%. The pattern is consistent with an early-warning tool: bias toward catching hard-to-spot positives while keeping the "safe" bucket highly trustworthy.

Overlap detection (logical-AND ensemble). To flag dual-role cases we require both base models to predict positive. Despite this stricter rule, the ensemble reached recall 93.8% (missed 2 of 32 known victim–perpetrators). The confusion figures were TN=76, FP=86, FN=2, TP=30, which imply precision 25.9%, specificity 46.9%, accuracy 54.6%, NPV 97.4%, F1 40.6%, and balanced accuracy 70.3%. Operationally, it cuts false positives sharply: of 763 unique alarms emitted by the two base models, only 86 survive the conjunction as false positives. Precision improves against a naïve "label-all-positive" baseline

(≈18.9%), and balanced accuracy exceeds that of either base model, reflecting better global discrimination while keeping recall very high.

~~5.~~ DISCUSSION

This study demonstrates that a dual-classifier framework can achieve high sensitivity in identifying adolescents at risk of simultaneously occupying victim and perpetrator roles. It also advances understanding of the victim–offender overlap by drawing on a representative sample of adolescents' self-reported experiences of multiple forms of victimization and involvement in diverse antisocial behaviors. Beyond informing the refinement of current theoretical models, the findings provide empirically grounded knowledge to support the development of effective, evidence-based prevention and intervention programs targeting adolescent victimization and perpetration (Yarkoni & Westfall, 2019).

network (perpetrator task) both achieve recall above 90 % (91.1 % and 90.0 %, respectively). In absolute terms this means that, on the held-out sets, the tree retrieves 339 of 372 victims and the network captures 199 of 221 perpetrators. Although precision is moderate (54.2 % for victims, 29.5 % for perpetrators), the deliberate bias toward recall is justified for an early-warning tool: the marginal cost of investigating a false alarm is lower than the social cost of overlooking a genuine case of violence towards a child or adolescent. Thus, in order to prevent the overlook of victim cases, the models prioritize protection, accepting more false positives to ensure that those at risk are not missed.

When combining the two high-recall classifiers through a logical AND, the ensemble identifies adolescents as dual-role only when both base models coincide. Despite the stricter criterion, recall remains very high (93.8%), with only 2 of the 32 known *victim–perpetrators* being missed. At the same time, the ensemble substantially reduces the number of false positives generated by each model individually: of the 763 unique alarms raised, only 86 are retained as *non-dual-role* after conjunction. This corresponds to an overall precision of 25.9%, which, while still limited, represents a notable improvement over the naïve baseline in which all adolescents would be labeled positive (precision = 18.9%).

The predictive value of negatives (NPV) emerges as a crucial strength of the framework. For victims, the NPV reached 74.2%, for perpetrators 91.8%, and for the overlap ensemble 97.4%. This means

Comentado [MOU47]: Creo que aqui falta motivar un poco mejor porque vemos que clasificar un victim (o perpetrator o both) como no victim es mucho peor que el caso contrario. Deberíamos escribir un pequeño párrafo explicando que el sistema no está pensado para tomar decisiones autónomamente (o automáticamente) si no en colaboración con expertos (psicólogos?) del centro escolar. De este modo, en todo momento el experto valida con su conocimiento las clasificaciones del sistema.

Comentado [I48]: Referencia colgante?

Comentado [MB49]: M'agrada molt aquest inici.

practitioners can rely on negative classifications with confidence, knowing that most unflagged adolescents are indeed at low risk. Such reliability enhances trust in the model as a safeguard, allowing scarce resources to be directed toward those flagged positive. By contrast, specificity was deliberately sacrificed in the base classifiers (24.9% for victims, 34.0% for perpetrators) in order to maximize sensitivity. While this design may strain frontline capacity, the ensemble helps mitigate the trade-off: by requiring both models to coincide, it raises specificity to 46.9% and reduces false positives to 86 out of 763 initial alarms, representing a substantial operational gain in efficiency.

~~The ensemble's balanced accuracy *(70.3 %)* underscores that the method does more than *"guess positive"*; its specificity *(46.9 %)* is nearly double that of the victim tree and *12* points higher than the perpetrator net. Qualitative inspection of the 86 false positives shows that most were either victims-only or perpetrators-only—profiles that still merit preventive attention even if they do not satisfy the strict dual-role definition.~~

Methodologically, the AND-ensemble demonstrates why decomposition outperforms direct modeling of the overlap under extreme imbalance. By splitting the task into two balanced problems, training remains stable, recall remains high, and interpretability is preserved. From an implementation perspective, this enables a two-tiered strategy: universal screening with the individual base models, followed by triage with the ensemble to concentrate resources. Such a framework is adaptable: thresholds can be adjusted to local prevalence, binary alarms can be scaled into tiered risk levels, and subgroup audits can monitor equity in errors. For generalization, external validation and continuous drift monitoring are essential steps before deployment.

Comentado [MB54]: Aquí no he modificado nada porque no acabo de entenderlo. Supongo que alguien que sepa del tema lo comprenderá.

Taken together, the findings suggest a two-tier deployment strategy. At the universal level, the victim and perpetrator classifiers can be applied as broad screening tools to ensure that almost no high-risk adolescent slips through undetected. In resource-constrained settings, however, practitioners may start directly with the ensemble rule, which prioritizes the subset most likely to occupy both roles and thus concentrates resources on the cases at greatest risk of mutual violence escalation. Within this framework, front-line screening can rely on the decision tree's handful of transparent questions to flag potential victims quickly, while secondary assessment benefits from the neural network's richer feature palette to refine perpetrator risk in borderline cases. Because both models converge on the same household-structure markers, intervention programmes should emphasize family-stability measures, supplemented with modules on impulse control, substance use, and digital consumption to more effectively prevent offending behavior.

This study demonstrates that a dual-classifier framework can reach very high recall in identifying adolescents at risk of simultaneously occupying victim and perpetrator roles, while also reducing the operational burden of false positives compared to single models. Prioritizing recall is justified within a cost–benefit logic of safety: overlooking a victim, a perpetrator, or both entails far greater social harm than investigating a false alarm. In this sense, the marginal cost of additional workload is outweighed by the security of not missing adolescents exposed to violence. Importantly, the system is not intended to function autonomously; it is conceived as a collaborative tool to assist school psychologists and frontline professionals, who validate and contextualize each flag with their expert judgment. This hybrid design ensures that recall maximization translates into protective practice rather than mechanical decision-making.

The predictive value of negatives (NPV) emerges as a crucial strength of the framework. For victims, the NPV reached ~~74.2~~91%, for perpetrators 93~~1.8~~%, and for the overlap ensemble ~~97.4~~93%. This means practitioners can rely on negative classifications with confidence, knowing that most unflagged adolescents are indeed at low risk. Such reliability enhances trust in the model as a safeguard, allowing scarce resources to be directed toward those flagged positive. By contrast, specificity was deliberately sacrificed in the base classifiers (~~24.9~~54% for victims, ~~34.0~~28% for perpetrators) in order to maximize sensitivity. While this design may strain frontline capacity, the ensemble helps mitigate the trade-off: by requiring both models to coincide, it ~~raises specificity to 46.9%~~maintains specificity at 23% and reduces false positives to ~~86~~60 out of 763 initial alarms, representing a substantial operational gain in efficiency.

~~The ensemble's balanced accuracy (70.3%) shows that it goes well beyond the naïve strategy of labeling all cases as positive. The fact that most false positives correspond to "victim-only" or "perpetrator-only" adolescents underscores its utility: even when not dual-role, these profiles remain relevant for preventive action. Crucially, the AND-ensemble does not simply inflate recall but introduces parsimony and precision, concentrating attention on cases with the greatest risk of escalation.~~

Several predictors deserve closer attention. Episodes of fuga (FUGAS.BN) act as accelerators of risky trajectories, bridging victimization and delinquency. Alcohol binge drinking and digital pornography exposure emerge as salient behavioral signals, aligning with prior research that associates them with impaired impulse control, sexual risk, and violence involvement. Variability in social support (APOYO.VAR) is as concerning as its absence, suggesting that unstable or inconsistent support networks can undermine resilience. Similarly, fluctuations in self-efficacy and impulse control may weaken protective factors, making adolescents more vulnerable. Gender, age, and sexual orientation effects also warrant careful interpretation: while females and minority groups are more often labeled as victims, this "majority-as-victim paradox" may reflect contextual or measurement biases, underscoring the need for cautious, non-stigmatizing use. Age-related patterns, where risks peak in mid-adolescence, further suggest critical windows for focused prevention.

Methodologically, the AND-ensemble demonstrates why decomposition outperforms direct modeling of the overlap under extreme imbalance. By splitting the task into two balanced problems, training remains stable, recall remains high, and interpretability is preserved. From an implementation perspective, this enables a two-tiered strategy: universal screening with the individual base models, followed by triage with the ensemble to concentrate resources. Such a framework is adaptable: thresholds can be adjusted to local prevalence, binary alarms can be scaled into tiered risk levels, and subgroup audits can monitor equity in errors. For generalization, external validation and continuous drift monitoring are essential steps before deployment.

Comentado [O55]: Sugerencias clave:
1.**Alto recall como objetivo de seguridad**: Explica explícitamente la lógica costo-beneficio: mayor carga de trabajo vs. evitar pasar por alto casos de riesgo.
2.**Importancia del NPV (Valor Predictivo Negativo)**: Destaca que la seguridad de los "negativos" fortalece la confianza en el modelo.
3.**Baja especificidad por diseño**: Discute cómo puede tensionar la capacidad de los sistemas de primera línea y la necesidad de priorizar recursos.
4.**Ganancias de balanced accuracy del AND-ensemble (70.3%)**: Señala que el modelo hace más que "etiquetar todo como positivo".
5.**Reducción de falsos positivos en el ensemble (86/763)**: Explica su relevancia operativa para seguimientos más eficientes.
6.**Parquedad vs. matiz**: Valora la complementariedad entre un árbol mínimo (víctimas) y una red compacta (perpetradores).
7.**Estabilidad y transportabilidad del split en PC_2 (0.54)**: Propón discutir sensibilidad a contexto/cohorte.
8.**Dominancia de estructura familiar (CONVIVEN.5, .2)**: Reforzar cómo converge en ambos modelos y vincular con literatura sobre inestabilidad.
9.**Interpretar la "inestabilidad familiar" como mecanismo de riesgo**.
10.**Paradoja de mayoría-víctima (ETNIA, PAÍS)**: Reconocerlo y discutir posibles explicaciones (contexto, muestreo, medición).
11.**Episodios de fuga (FUGAS.BN)**: Incorporar como acelerador transversal de trayectorias de riesgo.
12.**Alcohol en atracón y exposición digital (PORNO.T)**: Tratar como señales conductuales salientes y puntos de prevención.
13.**Apoyo social: varianza vs. nivel**: Discutir cómo la inestabilidad en apoyo puede ser tan problemática como su ausencia.
14.**Autoeficacia y control de impulsos**: Reflexionar sobre su papel protector vs. su volatilidad.
15.**Efectos de edad (negativos)**: Discutir ventanas de desarrollo para prevención focalizada.
...

Comentado [MB56]: No he entendido nada de esto. Le he pedido a chatgpt una traducción: Taken together, the findings support a two-step deployment strategy. First, the victim and perpetrator classifiers can be used as broad screening tools to ensure that very few high-risk adolescents go undetected. When resources are limited, practitioners may instead rely directly on the ensemble rule, which identifies the smaller subset of adolescents ...

Comentado [MB57]: He integrado esta discusión a la anterior.

Con formato: Fuente: 10 pto, Inglés (Estados Unidos), Resaltar

Con formato: Fuente: 10 pto, Inglés (Estados Unidos), Resaltar

Comentado [MB58]: Esto aparece explicado con otras palabras. SI esta parte está mejor, añadirla.

Con formato: Fuente: 10 pto, Inglés (Estados Unidos), Resaltar

In sum, the findings highlight that classifying a true victim or perpetrator as "negative" is far more detrimental than generating a false alarm. The deliberate prioritization of recall, coupled with the reassurance of high NPV and reduced false positives in the ensemble, provides a safety-first balance appropriate for preventive practice. By embedding the system within expert-guided processes, the model strengthens frontline capacity without replacing professional judgment. The dual-classifier framework thus offers both scientific insight into the victim–perpetrator overlap and a practical, adaptable tool for early detection and targeted intervention.

Comentado [O59]: Versión más enfocada a uso e interpretabilidad del resultado que nos proporcionan los modelos.

Comentado [I60R59]: Està molt ben explicat! Felicitats!

Con formato: Fuente: 10 pto, Inglés (Estados Unidos)

~~6. SUPPLEMENTAL MATERIALS~~

While the findings contribute valuable insights, several limitations should be noted. First, the dataset is derived from a single cross-sectional survey of school-attending adolescents. Although representative and relatively large, it may not adequately capture the experiences of young people outside the educational system or those from different cultural and regional contexts, which restricts the generalizability of the results. Second, the reliance solely on self-report measures introduces potential biases. Responses to sensitive items may be influenced by under- or over-reporting, social desirability, and recall inaccuracies, which could affect the observed associations between predictors and outcomes. Third, the marked imbalance in the victim–perpetrator category compared to other groups poses analytic challenges. Although the decomposition-and-ensemble strategy mitigates this issue by redistributing the task into better balanced binaries, residual imbalance remains and may limit the stability and calibration of the estimates. Finally, despite implementing safeguards such as cost-complexity pruning, cross-validation, regularization, and early stopping, the exploration of extensive hyperparameter grids carries a risk of overfitting the models to the specific characteristics of this dataset. These limitations should be considered when interpreting the findings and before applying the proposed classifiers in real-world screening contexts.

Comentado [NP61]: Buscar artículo self-report

Looking forward, several avenues for enrichment emerge. One important direction is the integration of the framework with educational and preventive policies. The parsimonious structure of the decision tree and the transparency of the neural network architecture render the approach well suited for embedding into school-based early warning systems. In such settings, the models could serve as initial filters to flag adolescents at elevated risk and connect them with psychosocial or family support resources, thereby improving the efficiency of preventive interventions. Another priority for future research is replicability across datasets. Independent validation using surveys from other countries, regions, or temporal cohorts is crucial to test the robustness and generalizability of the approach. Longitudinal data would also allow the exploration of causal pathways, clarifying how global predictors evolve over time and contribute to the dynamics of victimization and offending.

In conclusion, while the present study offers a transparent and recall-oriented framework for identifying dual-role adolescents, it is constrained by issues of representativeness, self-report bias, class imbalance, and potential overfitting. Future research should not only seek validation across diverse settings but also enrich the analytic strategy through cluster-based predictive modelling and qualitative interpretation of probabilistic outputs. Such extensions promise both theoretical advances in understanding the mechanisms of victim–perpetrator overlap and practical benefits for designing early screening tools and targeted interventions.

~~6.~~ SUPPLEMENTAL MATERIALS

~~6.2.~~ Tables

Table A1.

Binary variables

Variable	Value	Count	Percent
PAÍS	1	3432	91.1%
ETNIA.BN	1	751	19.9%
FUGAS.BN	1	519	13.8%
CONVIVEN.1	1	3468	92.1%
CONVIVEN.2	1	2841	75.4%
CONVIVEN.3	1	296	7.9%
CONVIVEN.4	1	114	3.0%
CONVIVEN.H~~5~~	1	2382	63.2%
CONVIVEN.0~~6~~	1	354	9.4%
GENERO_BIN_0	1	1805	47.9%
GENERO_BIN_1	1	1962	52.1%
ORIENTSEX.BN_1	1	3264	86.6%
ORIENTSEX.BN_2	1	503	13.4%

Table A2.

Ordinal variables

Variable	Value	Count	Percent
ABUSOSUBS1	0	1540	40.9%
ABUSOSUBS1	1	8	0.2%
ABUSOSUBS1	2	1183	31.4%
ABUSOSUBS1	3	668	17.7%
ABUSOSUBS1	4	347	9.2%
ABUSOSUBS1	5	21	0.6%
ABUSOSUBS2	0	754	20.0%
ABUSOSUBS2	1	1487	39.5%
ABUSOSUBS2	2	883	23.4%
ABUSOSUBS2	3	419	11.1%
ABUSOSUBS2	4	201	5.3%
ABUSOSUBS2	5	23	0.6%
PORNO.T	1	1791	47.5%
PORNO.T	2	727	19.3%
PORNO.T	3	362	9.6%
PORNO.T	4	482	12.8%
PORNO.T	5	405	10.8%

Table A3.

Continuous variables

Variable	Mean	SD	Median	Min	Max
EDAD	15.43	1.04	15.00	14.0	17.00
IMPULS.MEAN	2.40	0.35	2.38	1.0	4.00
IMPULS.MEDIAN	2.37	0.53	2.50	1.0	4.00
IMPULS.VAR	0.82	0.43	0.73	0.0	2.25
AUTOEFIC.MEAN	2.93	0.58	3.00	1.0	4.00
AUTOEFIC.VAR	0.44	0.39	0.24	0.0	2.16
APOYO.MEAN	3.27	0.58	3.43	1.0	4.00
APOYO.MEDIAN	3.46	0.76	4.00	1.0	4.00
APOYO.VAR	0.61	0.55	0.49	0.0	2.20
MORAL.MEAN	3.96	0.77	4.00	1.0	5.00
MORAL.VAR	0.62	0.67	0.40	0.0	3.84

Table 1.

List of global retained predictors after conceptual filtering

Feature	Brief description	Type
PAÍS	Country of birth	Binary (0~~1~~ = Spain, 1~~2~~ = Other)
ETNIA.BN	Binary ethnicity	Binary (0~~1~~ = ~~European~~Minority, 1~~0~~ = ~~minority~~European)
GENERO.BN	Binary gender	Binary (0 = male, 1 = female.)
ORIENTSEX.BN	Binary orientation	Binary (0~~1~~ = non-hetero, 1~~0~~ = h~~non-h~~etero.)
EDAD	Age in years	Ordinal (13,18)
FUGAS.BN	Has escaped	Binary (0~~1~~ = ~~Yes~~No, 1~~0~~ = ~~No~~Yes)
ABUSOSUBS1	Alcohol consumption freq.	Ordinal [1,5]
ABUSOSUBS2	Binge drinking (≥5 drinks)	Ordinal [1,5]

CONVIVEN1–7	Household co-residence flags	Categorical-Binary 7 binary questions
IMPULS1–8	Impulsivity scale items	Categorical-Ordinal 8 ordinal questions
AUTOEFIC1–5	Self-efficacy items	Categorical-Ordinal 5 ordinal questions
APOYO1–7	Social-support items	Categorical-Ordinal 7 ordinal questions
MORAL1–5	Moral-emotion items	Categorical-Ordinal 5 ordinal questions
PORNO.T	Combined frequency of pornography consumption	Ordinal [1,5]

Table 2.

Engineered Predictor Set Used for Modelling.

Predictor Label	**Description**	**Type**	**Descriptive Statistics**
PAÍS	Country of birth (0~~1~~ = Spain, 1~~2~~ = Other)	Binary	91.1% Spain
ETNIA.BN	Ethnicity: majority vs. minority	Binary	19.9% minority
EDAD	Age, clipped to the 14–17 range.	Continuous	M = 15.43, SD = 1.04
FUGAS.BN	Any runaway episode in the last 12 months	Binary	13.8% yes
ABUSOSUBS1	Alcohol use frequency (1–5; missing → 1)	Ordinal (0–5)	40.9% never; 31.4% monthly; 17.7% weekly; 9.2% ≥3/week; 0.6% daily
ABUSOSUBS2	Binge-drinking frequency (1–5; missing → 1)	Ordinal (0–5)	20.0% never; 39.5% monthly; 23.4% weekly; 11.1% ≥3/week; 0.6% daily
CONVIVEN.1	Lives with mother(s), biological or adoptive	Binary	92.1% yes
CONVIVEN.2	Lives with father(s), biological or adoptive	Binary	75.4% yes
CONVIVEN.3	Lives with mother's current partner	Binary	7.9% yes
CONVIVEN.4	Lives with father's current partner	Binary	3.0% yes
CONVIVEN.H~~5~~	Lives with siblings or step-siblings	Binary	63.2% yes
CONVIVEN.0~~6~~	Lives with other relatives (e.g., uncles, grandparents)	Binary	9.4% yes
CONVIVEN.7*	Lives in a residential center	Binary	<1% (merged with 0~~6~~ in model)
IMPULS.MEAN	Mean impulsivity score (reverse-coded)	Continuous	M = 2.40, SD = 0.35

Comentado [MB62]: He unido varias tablas en una sola. Las otras están en el Apéndice.

IMPULS.MEDIAN	Median impulsivity score	Continuous	M = 2.37, SD = 0.53
IMPULS.VAR	Variance of impulsivity responses	Continuous	M = 0.82, SD = 0.43
AUTOEFIC.MEAN	Mean self-efficacy (reverse-coded)	Continuous	M = 2.93, SD = 0.58
AUTOEFIC.VAR	Variance of self-efficacy responses	Continuous	M = 0.44, SD = 0.39
APOYO.MEAN	Mean perceived social support (reverse-coded)	Continuous	M = 3.27, SD = 0.58
APOYO.MEDIAN	Median social-support score	Continuous	M = 3.46, SD = 0.76
APOYO.VAR	Variance of social-support responses	Continuous	M = 0.61, SD = 0.55
MORAL.MEAN	Mean moral-emotion score (reverse-coded)	Continuous	M = 3.96, SD = 0.77
MORAL.VAR	Variance of moral-emotion responses	Continuous	M = 0.62, SD = 0.67
PORNO.T	Maximum reported frequency of pornography consumption	Ordinal (1–5)	47.5% never; 19.3% monthly; 9.6% weekly; 12.8% ≥3/week; 10.8% daily
GENERO_BIN_0	Male indicator	Binary	47.90%
GENERO_BIN_1	Female indicator	Binary	52.10%
ORIENTSEX.BN_1	Heterosexual indicator	Binary	86.60%
ORIENTSEX.BN_2	Non-heterosexual indicator	Binary	13.40%

Table 3.

Explained and cumulative variance by principal component, feature importance.

PC	Expl. var.	Cumul. var.	Direct var.	Direct corr.	Inv. var.	Inv. corr.
PC1	0.2198	0.2198	GENERO_BIN_0	0.6612	GENERO_BIN_1	-0.6612
PC2	0.1103	0.3300	CONVIVEN.H~~5~~	0.6375	ETNIA.BN	-0.2142
PC3	0.0903	0.4204	ORIENTSEX.BN_2	0.6174	ORIENTSEX.BN_1	-0.6174
PC4	0.0842	0.5046	CONVIVEN.H~~5~~	0.6152	CONVIVEN.2	-0.2919
PC5	0.0679	0.5725	EDAD	0.4543	CONVIVEN.2	-0.3091
PC6	0.0576	0.6300	CONVIVEN.2	0.5891	CONVIVEN.3	-0.2501
PC7	0.0486	0.6786	FUGAS.BN	0.5380	EDAD	-0.5117

Comentado [MB63]: Esta tabla igual se podría poner en resultados, porque qué componentes concentran la mayor parte de la varianza es parte de lo que hemos descubierto ¿no?

Comentado [O64R63]: Sí que podría ponerse en resultados y de hecho con un mapa de temperatura que es más visual que los propios valores.

Comentado [MOU65R63]: Si, otra imagen que valdría más que mil tablas.

PC8	0.0402	0.7188	APOYO.MEDIAN	0.4604	EDAD	-0.3984
PC9	0.0337	0.7525	CONVIVEN.0~~6~~	0.8586	CONVIVEN.1	-0.3753
PC10	0.0320	0.7845	PORNO.T	0.7411	FUGAS.BN	-0.3696
PC11	0.0289	0.8133	CONVIVEN.1	0.6759	CONVIVEN.3	-0.5137
PC12	0.0248	0.8382	ABUSOSUBS1	0.5635	PORNO.T	-0.4473
PC13	0.0227	0.8609	PAÍS	0.8077	ETNIA.BN	-0.3673
PC14	0.0213	0.8821	CONVIVEN.3	0.6095	PAÍS	-0.3597
PC15	0.0200	0.9021	APOYO.VAR	0.5633	IMPULS.MEDIAN	-0.3460
PC16	0.0176	0.9197	MORAL.MEAN	0.5787	IMPULS.MEDIAN	-0.4630
PC17	0.0165	0.9362	AUTOEFIC.MEAN	0.4594	AUTOEFIC.VAR	-0.4345
PC18	0.0142	0.9503	APOYO.VAR	0.5400	IMPULS.VAR	-0.4925
PC19	0.0135	0.9638	IMPULS.VAR	0.7165	AUTOEFIC.VAR	-0.2834
PC20	0.0097	0.9735	ABUSOSUBS2	0.7991	ABUSOSUBS1	-0.5722
PC21	0.0094	0.9829	CONVIVEN.4	0.9313	CONVIVEN.3	-0.2522
PC22	0.0079	0.9908	AUTOEFIC.MEAN	0.6398	IMPULS.VAR	-0.2323
PC23	0.0065	0.9973	MORAL.VAR	0.7459	AUTOEFIC.VAR	-0.2182
PC24	0.0016	0.9989	APOYO.MEAN	0.8379	APOYO.MEDIAN	-0.4792
PC25	0.0011	1.0000	IMPULS.MEAN	0.8527	IMPULS.MEDIAN	-0.5137
PC26	0.0000	1.0000	GENERO_BIN_0	0.7071	IMPULS.MEAN	-0.0000
PC27	0.0000	1.0000	ORIENTSEX.BN_2	0.7071	GENERO_BIN_0	-0.0005

Comentado [O66]: Propongo poner todas las tablas como material adjunto y referenciarlas en el estudio directamente.

Table 4.

Unified over-fitting control and regularisation across classifiers

Phase	Purpose	Model	Mechanism
Stratified split	Preserve class balance;	Both	Train/test and val split stratified
Depth (1–20)	Diagnose variance	tree	Structural regularisation by maximum depth
CCP	Generate candidate trees;	tree	Prune with cost–complexity sequence
α-grid search	Optimise recall	tree	Select $\alpha\star$=0.01495
Class weighting	Emphasise minority class	NN	Loss re-weighting
Dropout layers	Neuron co-adaptation	NN	Apply dropout p=0.10
Early stopping	Halt training	NN	Stop **5** epochs without recall improvement
Gap detector	Against over-fitting spikes	NN	Stop if train–validation recall gap >0.10
Grid search	Find simplest architecture	NN	Explore widths/regularisation
Artefact export	Ensure reproducibility	Both	Persist final model objects

Table 5.

Interpretation of evaluation metrics where T = true, F = false, P = positive, N = negative.

Metric	Formula	Description	Utility in this study
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Overall fraction of correct predictions	Can be misleading under class imbalance; Used only as a rough check.
Precision	$\frac{TP}{TP+FP}$	Share of positive calls that are truly positive	Reflects investigation workload: lower precision → more false alarms frontline teams must review.
Recall	$\frac{TP}{TP+FN}$	Share of true positives that are detected	Missing a victim or perpetrator imposes high social cost.
F1	$2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$	Harmonic mean of precision and recall	Convenient single number for model ranking when recall and precision both matters.

Specificity	$\frac{TN}{TN + FP}$	Share of true negatives correctly dismissed	Indicates how many low-risk youths are correctly left unflagged, balancing recall.
NPV	$\frac{TN}{TN + FN}$	Share of negative calls that are truly negative	Confidence that a "safe" label indeed, corresponds to low risk.
Balanced Accuracy	$\frac{Recall + Specificity}{2}$	Mean of recall and specificity	Adjusts for imbalance; gauges discrimination while keeping recall central.

Table 6.

Confusion matrix for the victim-detection tree

	Pred. Non-Victim	**Pred. Victim**
Actual Non-Victim	TN = 95	FP = 287
Actual Victim	FN = 33	TP = 339

Table 7.

Confusion matrix for the perpetrator-detection network

	Pred. Non-Perp.	**Pred. Perp.**
Actual Non-Perp.	TN = 245	FP = 476
Actual Perp.	FN = 22	TP = 199

Table 8.

Confusion Matrix of the Ensembled Victim–Perpetrator Classifier

	Predicted 0	**Predicted 1**	**Total Actual**
Actual 0	TN = 76	FP = 86	162
Actual 1	FN = 2	TP = 30	32

Table 9.

Comparative Performance Metrics of the Victim, Perpetrator, and Ensemble Classifiers

Metric	**Victim Classifier Performance (%)**	**Perpetrator Classifier Performance (%)**	**Overlap Classifier Performance (%)**
Accuracy	~~0.576 (57.6)~~58%	~~0.471 (47.1)~~43%	~~0.546 (54.6)~~41%
Precision	~~0.542 (54.2)~~54%	~~0.295 (29.5)~~28%	23%~~0.259 (25.9)~~

Recall	~~0.911 (91.1)~~91%	~~0.900 (90.0)~~93%	~~0.938 (93.8)~~93%
F1-score	~~0.678 (67.8)~~68%	~~0.445 (44.5)~~45%	~~0.406 (40.6)~~40%
Specificity	~~0.249 (24.9)~~24%	~~0.340 (34.0)~~32%	~~0.469 (46.9)~~46%
NPV	~~0.742 (74.2)~~91%	~~0.918 (91.8)~~92%	~~0.974 (97.4)~~97%
Balanced Accuracy	~~0.580 (58.0)~~58%	~~0.620 (62.0)~~62%	~~0.703 (70.3)~~70%

Comentado [MB67]: Revisar que los porcentajes sean correctos

Comentado [O68R67]: Corregido

Table 10.

Raw-Space Feature Coefficients Underlying the Decision-Tree Split on PC_2

#	Predictor	β_j	Weight	Effect
1	**CONVIVEN.H~~5~~**	0.637478	0.229	+
2	**CONVIVEN.2**	0.546152	0.196	+
3	**ETNIA.BN**	-0.214185	0.077	–
4	**ORIENTSEX.BN_1**	0.203226	0.073	+
5	**ORIENTSEX.BN_2**	-0.203226	0.073	–
6	**FUGAS.BN**	-0.172597	0.062	–
7	**CONVIVEN.1**	0.156661	0.056	+
8	**PAÍS**	-0.130422	0.047	–
9	**CONVIVEN.3**	-0.122660	0.044	–
10	**CONVIVEN.06**	-0.106702	0.038	–
11	**APOYO.MEDIAN**	0.040205	0.014	+
12	**APOYO.VAR**	-0.045495	0.016	–
13	**EDAD**	-0.031849	0.011	–
14	**ABUSOSUBS1**	-0.017070	0.006	–
15	**ABUSOSUBS2**	-0.013711	0.005	–
16	**CONVIVEN.4**	-0.013151	0.005	–
17	**AUTOEFIC.MEAN**	0.018166	0.007	+
18	**AUTOEFIC.VAR**	-0.021052	0.008	–

19	**APOYO.MEAN**	0.034071	0.012	+
20	**PORNO.T**	-0.016569	0.006	–
21	**IMPULS.MEDIAN**	0.000463	0.000	+
22	**IMPULS.MEAN**	-0.000006	0.000	–
23	**IMPULS.VAR**	-0.005535	0.002	–
24	**MORAL.MEAN**	0.010998	0.004	+
25	**MORAL.VAR**	-0.008597	0.003	–
26	**GENERO_BIN_0**	0.008369	0.003	+
27	**GENERO_BIN_1**	-0.008369	0.003	–

Note. Raw-space coefficients that reproduce the decision-tree rule $PC_2 \leq t$. "Weight" is the normalized absolute value; "Effect" shows whether an increase in the variable moves a respondent towards (+) or away from (–) the victim class.

Table 11.

Perpetrator neural network SHAP importance of principal components

COMPONENT	IMPORTANCE (%)
PC2	18.69
PC4	14.06
PC5	10.01
PC7	9.27
PC1	8.73
PC6	8.60
PC16	5.65
PC10	5.46
PC21	5.26
PC17	4.02
PC9	3.54

PC18	2.68
PC22	1.53
PC11	1.41
PC3	0.44
PC20	0.39
PC14	0.15
PC19	0.08
PC18	0.02
PC15	0.01
PC13	0.002
PC12	0.001

Table 12.

Perpetrator neural network SHAP-derived importance of engineered predictors

FEATURE	**ψ_i**	**WEIGHT**	**EFFECT**
CONVIVEN.H~~5~~	22.02	0.201	+
CONVIVEN.2	9.46	0.086	+
FUGAS.BN	9.02	0.082	+
ORIENTSEX.BN_2	-7.74	0.071	–
ORIENTSEX.BN_1	7.74	0.071	+
ABUSOSUBS2	7.34	0.067	+
APOYO.VAR	6.92	0.063	+
PORNO.T	6.35	0.058	+
GENERO_BIN_1	-4.45	0.041	–
GENERO_BIN_0	4.45	0.041	+
CONVIVEN.0~~6~~	4.19	0.038	+
AUTOEFIC.MEAN	3.33	0.030	+

Con formato: Sangría: Izquierda: 0 cm, Sangría francesa: 1,27 cm

IMPULS.MEDIAN	-2.65	0.024	–
EDAD	-2.40	0.022	–
ETNIA.BN	1.98	0.018	+
IMPULS.MEAN	-1.71	0.016	–
CONVIVEN.3	-1.28	0.012	–
APOYO.MEAN	-1.10	0.010	–
AUTOEFIC.VAR	1.07	0.010	+
CONVIVEN.4	1.00	0.009	+
ABUSOSUBS1	0.83	0.008	+
APOYO.MEDIAN	0.81	0.007	+
MORAL.VAR	0.66	0.006	+
PAÍS	0.64	0.006	+
IMPULS.VAR	-0.57	0.005	–
CONVIVEN.1	-0.09	0.001	–
MORAL.MEAN	0.03	0.000	+

Table 13.

Ensemble feature-importance breakdown with condensed column names

Feature	PW	Dir	w_DT	d_DT	w_NN	d_NN
CONVIVEN.H~~5~~	0.262997	+	0.229	+	0.201	+
CONVIVEN.2	0.172477	+	0.196	+	0.086	+
ORIENTSEX.BN_2	0.088073	–	0.073	–	0.071	–
ORIENTSEX.BN_1	0.088073	+	0.073	+	0.071	+
ABUSOSUBS2	0.037920	+	0.005	–	0.067	+
ETNIA.BN	0.036086	–	0.077	–	0.018	+
CONVIVEN.3	0.034251	–	0.044	–	0.012	–
CONVIVEN.1	0.033639	+	0.056	+	0.001	–

PORNO.T	0.031804	+	0.006	–	0.058	+
APOYO.VAR	0.028746	+	0.016	–	0.063	+
GENERO_BIN_1	0.026911	–	0.003	–	0.041	–
GENERO_BIN_0	0.026911	+	0.003	+	0.041	+
PAÍS	0.025076	–	0.047	–	0.006	+
AUTOEFIC.MEAN	0.022630	+	0.007	+	0.030	+
EDAD	0.020183	–	0.011	–	0.022	–
IMPULS.MEDIAN	0.014679	–	0.000	+	0.024	–
APOYO.MEDIAN	0.012844	+	0.014	+	0.007	+
FUGAS.BN	0.012232	+	0.062	–	0.082	+
IMPULS.MEAN	0.009786	–	0.000	–	0.016	–
IMPULS.VAR	0.004281	–	0.002	–	0.005	–
MORAL.MEAN	0.002446	+	0.004	+	0.000	+
CONVIVEN.4	0.002446	+	0.005	–	0.009	+
MORAL.VAR	0.001835	+	0.003	–	0.006	+
AUTOEFIC.VAR	0.001223	+	0.008	–	0.010	+
APOYO.MEAN	0.001223	+	0.012	+	0.010	–
ABUSOSUBS1	0.001223	+	0.006	–	0.008	+
CONVIVEN.0~~6~~	0.000000	+	0.038	–	0.038	+

6.3. Equations

$$eq.1: Scale - level mean, median and variance$$

$$\mu = \frac{1}{m}\sum_{i=1}^{m} x_i, \tilde{x} = median\{x_i\}, \sigma^2 = \frac{1}{m}\sum_{i=1}^{m}(x_i - \mu)^2$$

$$eq.2: Min - Max\ scaling\ and\ mean - centering$$

$$x_{ij}^{scaled} = \frac{x_{ij} - x_{ij}}{x_{ij} - x_{ij}}, \widetilde{x_{ij}} = x_{ij}^{scaled} - \frac{1}{n}\sum_{i}^{n} x_{ij}^{scaled}$$

$$eq.3: Dimensionality\ reduction\ (PCA)$$

$$C = \frac{1}{n-1}X_{cent}^{\top}X_{cent},\ Cv_k = \lambda_k v_k\ (\lambda_1 \geq \cdots \geq \lambda_{27} \geq 0),\ z_{ik} = x_i^{(cent)} \cdot v_k,\ (k = 1, \dots, 27)$$

$$eq\ (4): Cost - complexity\ criterion\ R\alpha(T)$$

$$R\alpha(T) = R(T) + \alpha\,|T|$$

where *R(T)* is the empirical error, |*T*| the number of leaves, and $\alpha \geq 0$ the complexity penalty.

$$eq\ (5): Forward - pass\ equations\ of\ the\ neural\ network$$

$$a_1 = \sigma_1(W_1 z + b_1), \quad |a_1| = 64$$

$$\widetilde{a_1} = Drop_{0.10}(a_1), \quad (training\ only)$$

$$a_2 = \sigma_2(W_2\widetilde{a_1} + b_2), \quad |a_2| = 8$$

$$\hat{y} = \sigma_3(w_3^{\top} a_2 + b_3), \quad \hat{y} \in [0,1]$$

where σ_1 is the identity *(linear),* σ_2 and σ_3 are element-wise *sigmoids, and* $Drop_p$ denotes dropout with keep-probability *1-p*. *eq (5)* encapsulates the full forward pass referenced in **Table 4**.

$$eq\ (6): Victim - Perpetrator\ conjuntion\ equation.$$

$$\hat{y}vp = \mathbf{1}\{\hat{y}\mathrm{v} = 1\} \wedge \mathbf{1}\{\hat{y}\mathrm{p} = 1\}$$

$$eq\ (7): Inequality\ after\ reversing\ PCA$$

$$w_2^{\top}(z - \mu) \leq t$$

$$eq\ (8): Hyperplane\ in\ raw\ feature\ space$$

$$z = S(x - d_{min})\ and\ regroup\ constants: \quad \beta^{\top}x \leq c$$

$$eq\ (9): Normalised feature - importance weights$$

$$w_j = \frac{|\beta_j|}{\sum_{k=1}^{27}|\beta_k|}, \quad j = 1, \dots, 27$$

$$eq\ (10){:}\ Let\ M\ be\ the\ number\ of\ input\ principal\ components\ (PC).$$

$$and\ f(x)\ the\ network\ output\ for\ a\ sample\ x.$$

$$The\ SHAP\ value\ of\ component\ k\ for\ x\ is\ the\ Shapley\ value.$$

$$\varphi_k(x) = \sum_{S\subseteq M\setminus\{k\}} \frac{|S|!\ (M-|S|-1)!}{M!}\left[f_{S\cup\{k\}}\left(x_{S\cup\{k\}}\right) - f_S(x_S)\right]$$

$$Where\ f_s(x_s)\ is\ the\ prediction\ when\ only\ the\ feature\ in\ subset\ S\ are\ present.$$

$$eq(11){:}\ For\ N\ observations\ ,the\ global\ (mean-absolute)\ importance\ of\ PCk\ is$$

$$I_k = \frac{1}{N}\sum_{j=1}^{N}\left|\varphi_k!\left(x^{(j)}\right)\right|$$

$$which\ we\ convert\ to\ a\ percentage\ of\ total\ importance\ via\ eq.12$$

$$eq\ (12){:}\ Normalised\ importance\ of\ PC\ k$$

$$\widetilde{I_k} = 100\cdot\frac{I_k}{\sum_l I_l}\quad (\%\ of\ total)$$

$eq(13){:}\ Each\ engineered\ predictor\ x_i\ is\ a\ lineal\ combination\ of$ PCs,

$where\ W = (w_{k,i})$ is the loading matrix.

$$We\ distribute\ the\ SHAP\ importance\ of\ every\ PC\ onto\ the\ original\ variables{:}$$

$$\psi_i = \sum_k w_{k,i}\, I_k\,, \qquad \Psi_i = 100\cdot\frac{|\psi_i|}{\sum_j|\psi_j|}$$

$$where\ \psi_i\ is\ the\ signed\ contribution\ and\ \Psi_i\ its\ percentage\ share\ of\ the\ total.$$

****Más info que he resumido, se puede dejar o quitar a vuestro criterio

The modelling strategy pursued two complementary goals: (i) maximize sensitivity to both victimization and offending *(high recall for the positive class)* and (ii) keep model complexity low enough for frontline interpretability. All model training processes were conducted in collaboration with the eAlicia AI research team.

3.4.1. Victim Classifier

Model and rationale. A decision tree classifier was trained on the first K=18 principal components, which together explain approximately 95% of the variance. To maximize interpretability while controlling variance, the tree was pruned using cost–complexity pruning **(eq.4)**. The resulting model is extremely sparse—a depth-1 stump—providing transparent, rule-based predictions (**Fig. 1**).

Model selection and tuning. A grid search over the pruning parameter identified the optimal value $\alpha * = 0.01495$. This choice retains a single split on the second principal component (PC_2) at a threshold of 0.54, yielding a minimal model without sacrificing recall.

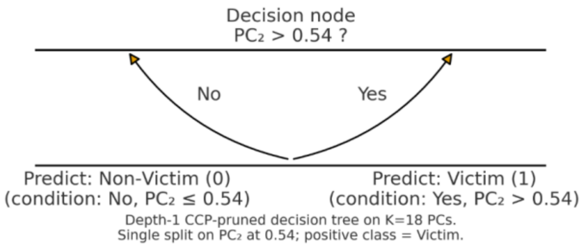
Decision rule. The classifier comprises a single thresholding rule on PC_2: observations are assigned according to the side of the split relative to 0.54 (terminal-node class labels determined by in-sample majority).

Performance. On the hold-out set, the depth-1 model attains a recall of 0.91 for the victim class, combining high sensitivity with maximal transparency.

Fig.1.

Pruned decision Tree

Comentado [MOU69]: Ejemplo de que vale más una imagen que mil palabras.



3.4.2. Perpetrator Classifier

Model and rationale. A compact feed-forward neural network was trained on the first 22 principal components, which together explain approximately 99% of the variance. The model comprises roughly 2,000 trainable parameters, balancing expressive capacity with generalization, promoting stable optimization, and preserving interpretability while capturing relevant non-linear associations with perpetration.

Comentado [MOU70]: Referenciadlo

Comentado [O71R70]: Wythoff, B. J. (1993). Backpropagation neural networks: A tutorial. *Chemometrics and Intelligent Laboratory Systems, 18*(2), 115–155. https://doi.org/10.1016/0169-7439(93)80052-J

Architecture. Let $z \in R^{22}$ denote the PCA (Principal Component Analysis) input vector. The network consists of a hidden layer of 64 units with a linear activation, followed—during training only—by 10% dropout; a second hidden layer of 8 units with a sigmoid activation; and a single sigmoid output unit. The scalar output $\hat{y} \in [0,1]$ is interpreted as the estimated probability of belonging to the perpetrator (positive) class. The complete forward pass is specified in **(eq.5)**.

Regularization and tuning. Class imbalance and overfitting were addressed via inverse-prevalence class weighting, dropout, early stopping on validation recall, and a custom overfit-gap check. Hyperparameters were selected through grid search, prioritizing the simplest architecture that satisfied the target recall **(Table 4)**.

Performance. On the hold-out set, the classifier achieved a recall of 0.900 for the perpetrator class.

3.4.3. Overlap Classifier

Model and rationale. Dual-role (victim–perpetrator) cases are identified via a smart ~~n~~ AND-ensemble ~~that combines the base learners~~that combines information on both roles. A compact feed-forward neural is trained to perform the ensemble operation. Basically, the ensemble acts as an AND logical operation, with the freedom to change the result provided if one of the cases (e.g., perpetrator or victim) are near the decision boundary. A case is flagged as overlap only when both the victim classifier and the perpetrator classifier return positive decisions. This conjunction yields a recall-oriented yet parsimonious detector while preserving full transparency.

Construction. Let $\hat{y}pv, \hat{y}p \in \{0,1\}$ denote the binary outputs of the victim decision tree **(eq.4)** and the perpetrator neural network **(eq.5)**, respectively. The overlap prediction is defined by the logical conjunction **(eq.6)**

$$\hat{y}vp = \mathbf{1}\{\hat{y}v = 1\} \wedge \mathbf{1}\{\hat{y}p = 1\}$$

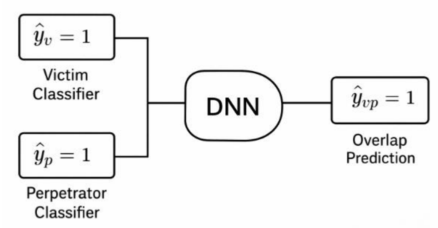
This rule triggers only when both base models flag a case positive.

Fig.2.

Comentado [MOU72]: Añadid referencia

Comentado [O73R72]: Gewers, F. L., Ferreira, G. R., de Arruda, H. F., Silva, F. N., Comin, C. H., Amancio, D. R., & da F. Costa, L. (2021). Principal component analysis: A natural approach to data exploration. *ACM Computing Surveys, 54*(4), Article 70. https://doi.org/10.1145/3447755

Comentado [MOU74]: ¿Podemos poner una imagen sencilla que lo explique, como la de arriba?



Overlap Classifier via an smart AND-ensemble

Justification. A direct model trained on the overlap label y_{vp} would face extreme class imbalance, leading to unstable optimization and potential loss of sensitivity. Decomposing the task into two better-balanced problems—victim and perpetrator detection—and combining them via conjunction preserves sensitivity, stabilizes training, and maintains an interpretable decision rule appropriate for frontline use.

Model Explainability (XAI)

3.5.1. Decision-Tree Feature Importance

Overview. The cost–complexity–pruned victim tree reduces to a single internal split in principal-component space, with decision boundary specified in **Fig 1.**; model behaviour is therefore governed by a threshold on the second principal component (PC_2). Raw-feature contributions are recovered by an algebraic back-projection from PC space to the original variables in three steps: (i) re-express the PC split in terms of rotated, centred features **(eq .7)**; (ii) invert the preprocessing to un-scale variables via the min–max transform and regroup constants **(eq.8)**; and (iii) convert the resulting raw-space coefficients into unit-free importances by normalising absolute values to sum to one **(eq.9)**.

Reporting and interpretation. The procedure yields, for each original predictor, a signed coefficient, a normalised weight, and a direction of effect ("+" if larger values move a case toward the victim class; "–" otherwise). These results are reported in **Table 10**, which lists the raw-space coefficients, their normalised weights, and qualitative effects.

3.5.2. Neural-Network Feature Importance

Overview. Feature importance for the perpetrator classifier is derived in two stages to preserve the network's non-linear scoring while enabling interpretation at the level of the original predictors. Following: (i) **Importance in PC space.** Shapley values are computed for each input principal component relative to the network output **(eq.10)**. Absolute Shapley magnitudes are aggregated across observations to obtain global importances **(eq.11)**, then normalised to percentages **(eq.12)**. (ii) **Propagation to original predictors.** Because each engineered predictor is a linear combination of PCs, the PC-level importances are distributed back to the raw variables via the loading matrix, producing a signed contribution per predictor and a normalised percentage share as defined in **(eq.13)**.

Reporting and interpretation. The global ranking of principal components is presented in **Table 11.** The propagated variable-level importances and directions (toward or away from the perpetrator class) are presented in **Table 12**. Together, these summaries link the network's latent representation to interpretable effects on the original predictors.

3.5.3. Ensemble (Overlap) Model Feature Importance

Overview. The overlap detector flags a case as dual-role only when both base models are positive, combining the victim decision tree **(eq.3)** and the perpetrator neural network **(eq.5)** through a ~~logical~~smart logical conjunction defined in **Table 5**. Per-feature ensemble importance is obtained by reconciling the tree and network importances. For features present in both models, directions of effect are first compared; when directions agree, their weights are summed and the shared direction is retained, and when directions disagree, the smaller weight is subtracted from the larger, with the sign of the result determining the net direction. Absolute values are then re-scaled so that the combined weights sum to one.

Reporting and interpretation. The full per-feature breakdown—showing combined ensemble weight and net direction alongside the constituent tree and network contributions—is provided in **Table 13**. This approach preserves the transparency of the depth-1 tree, respects the broader signal distribution of the neural network, and yields an interpretable account of which global predictors most strongly drive dual-role flags under the conjunction rule.

Comentado [O75]: He incluido “Model Explainability” dentro de la metodología porque es lo que nos permite extraer conclusiones victimológicas. No obstante, podríamos moverlo a *Supplemental Materials* para aligerar el cuerpo del paper y recortar unas páginas. ¿Qué os parece?

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